Module 1: What is Data Science?

DSC 210 Foundations of Data Science

References: - Hands-on Introduction to Data Science with Python (CC BY-NC-SA 4.0)

Last major revision: 2026-08-02

Key Concepts

- State a working definition of data science and apply it to borderline cases
- Describe data science as the intersection of Math & Statistics, Hacking Skills, and Substantive Expertise
- Apply the 5-C checklist to evaluate the ethical dimensions of a data science project

Activity 0.1 - Is this data science? (warm-up)

The definition of data science is fuzzy by design. Below is a list of tasks. For each one, decide as a group:

> Is this data science, statistics, software engineering, or some combination?

#	Task
1	A/B testing two versions of a website button
2	Computing average monthly sales for a store
3	Training a spam filter on 10 million emails
4	Making a bar chart in Excel
5	Running a randomized clinical trial for a new drug
6	Building a facial recognition system for phone unlocking
7	Recommending which Netflix show you should watch next
8	Writing a Python script to rename 10,000 files
9	Surveying students about campus dining and reporting percentages

1. Definition of Data Science

Is there a pattern to the tasks your group called "definitely data science"? What feature makes the label fit?

Data science often combine *learning from data* (not just computing a statistic), *scale* (computational work with large or complex datasets), and producing and communicating insights or predictions that can inform decisions or actions. But the boundary is genuinely blurry, and arguing about it is productive.

Definition of Data Science (for now): > *Data science is the art (or practice) of gaining, interpreting, and communicating insights from complex data through digital techniques.*

1.1 The three circles

One especially common way to illustrate data science is with a Venn diagram: the overlap between Hacking Skills, Math & Statistics Knowledge, and Substantive Expertise. The diagram is Drew Conway's, first published in 2010, and it has shaped how the field describes itself ever since.

What are "Hacking Skills"? In Conway's words: "Data is a commodity traded electronically; therefore, in order to be in this market you need to speak hacker. This, however, does not require a background in computer science. Being able to manipulate text files at the command-line, understanding vectorized operations, thinking algorithmically; these are the hacking skills that make for a successful data hacker."

Activity 1.1 - What are your strengths?

Consider the three categories:

- **Math & Statistics:** probability, inference, modeling
- **Hacking Skills:** programming, databases, algorithms
- **Substantive Expertise:** a substantive field you know well (biology, economics, sports, video games, …)

Part A. Take a moment and rate yourself on a scale from 0 (no background) to 10 (expert) in each of the three categories.

Part B. Group up with 2 other students and treat your group as a single team. Where are the strengths of your team? Where are the gaps? If you were hiring one more person, what would their profile look like?

Part C. Do you agree that no single person masters all three circles equally? Is there a risk in specializing too much in one circle?

1.2 It's a big and beautiful world

Data science isn't just about dealing with numbers or coding! - Leveraging digital tools and statistical methods to draw insights from data - Applying these insights to a specific context or domain.

No single person can master all facets of data science!

2. Why Now?

1. Exponential increase in volume of data generated
2. Expansion of statistical methodologies
3. Greater data-handling capabilities
4. Significant progress in algorithm development
5. Revolutions in data visualization

Concurrent Developments Leading to Data Science

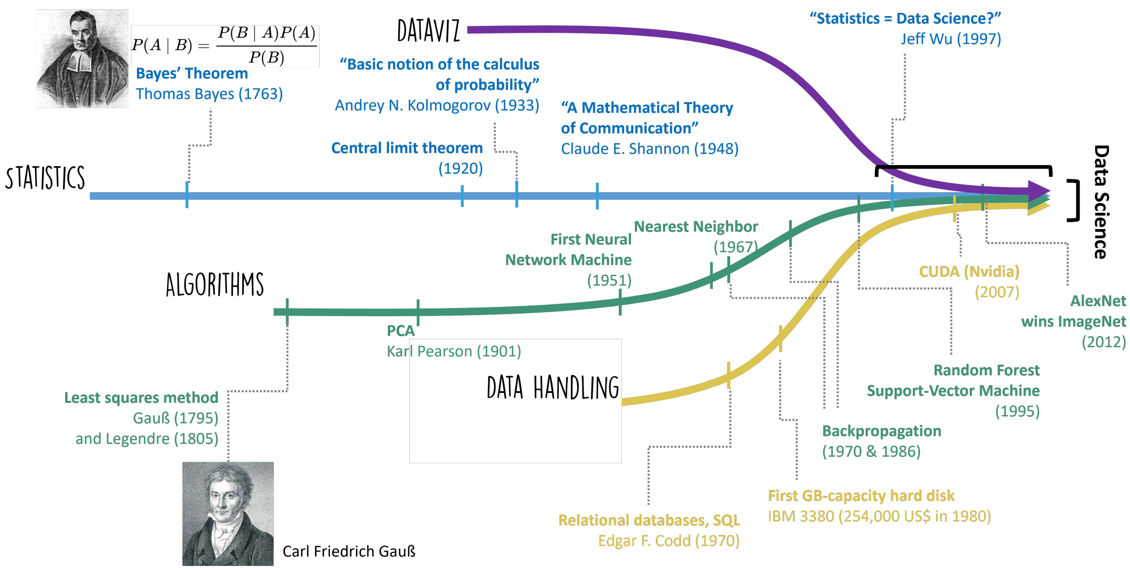

Figure 1: Concurrent Developments Leading to Data Science

2.1 How do you learn Data Science?

Here are a few ideas:

- *Practice* the concepts! Just like with the fields Data Science draws from, the best way to learn Data Science is by *doing* it!
- Use technology, but don't rely on it!
- Coding is usually the *simple* part! The most important skills in Data Science center around effective communication.

3. Ethical Questions

Most data science tasks bring ethical questions and societal implications - analyzing online user behavior - studying internal corporate processes - evaluating political campaigns - communicating scientific content

Class Exercise 1.1 - Content recommendations and the 5-C checklist

Scenario. A major video platform recommends what you watch next. The algorithm is trained to maximize the total time users spend watching. It has access to your age, location, watch history, search history, device type, and time of day. The algorithm is not explained to users and is updated continuously.

Step 1. Apply the 5-C checklist below to assess this system. For each criterion, write one sentence of assessment and assign a risk level (Low / Medium / High) in the table below.

The five criteria:

- **Consent:** Did users meaningfully agree to this use of their data?
- **Clarity:** Is it transparent what is collected and why?
- **Consistency:** Are all users treated alike across demographic groups?
- **Control and Transparency:** Can users see or correct their profile?
- **Consequences and Harm:** Who could be hurt, and how seriously?

Criterion	Your assessment	Risk
Consent		
Clarity		
Consistency		
Control and Transparency		
Consequences and Harm		

Step 2. Discuss the following three questions as a group:

1. The platform might argue that users can turn off recommendations. Does opt-out satisfy the Consent criterion? Why or why not?
2. Some research suggests recommendation algorithms can amplify extreme content because it generates longer watch sessions. Which 5-C criterion does this most directly violate?
3. Is the algorithm itself the problem, or is the objective function (maximize watch time) the problem? Does the distinction matter for assigning accountability?

Activity 1.2 - Your data

Ethical issues arise whenever data is collected, categorized, analyzed, and used to influence people. Let's make this personal.

Part A. Brainstorm: what data is being collected about you *right now* or *in the last 24 hours*? Include:

- Apps and services (phone, laptop, streaming, banking, social)
- Physical sensors (fitness trackers, building cameras, card swipe logs)
- Institutional records (the university, your health insurer)

Aim for at least 10 distinct data streams.

Part B. For three of those streams, answer:

- Is this data that could have been collected 30 years ago (No internet, no cell phones, etc.)
- Who stores the data?
- What could be *inferred* from it that you didn't directly share?
- Would you be comfortable if a stranger had the same access?

Part C. Data science is "never only technical, it always involves choices." What choices did someone make in designing one of your data streams? What alternative choices were available?

Closing question: After doing this inventory, does your overall comfort level with the data being collected about you change? What specifically shifts?

Note: The gap between what is *collected* and what can be *inferred* is a deep issue.

- Location data alone can reveal medical conditions (repeated visits to a specialty clinic), religious practice (visits to places of worship), political activity (attendance at rallies), and relationship status.

Class Example 1.2 - Targeted marketing

Suppose you're a marketing analyst for a large chain of department stores. You've identified a pretty interesting pattern using customer purchasing data.

If a combination of items is purchased, customers tend to purchase maternity clothes between 3-6 months later.

These items include: - Unscented lotion - Supplements (calcium, magnesium, etc.) - Cotton balls (large quantities) - Washcloths - Large purse

Your company is able to associate purchases across dates using unique "Guest IDs", which incorporate credit card information, name, email, location, app usage, and other online activity.

> Consider a shopper named Jenny Ward, who is 23, lives in Atlanta and in March bought cocoa-butter lotion and a purse large enough to double as a diaper bag in-store, and zinc and magnesium supplements and a bright blue rug online.

Your company's ultimate goal is to sell more products. In particular, your immediate goal is to sell more maternity clothes.

Small group questions:

1. From the perspective of a marketing analyst, how would you prioritize customers for targeted advertising? Is there additional information needed to increase the efficacy of this marketing campaign?
2. How does the customer (say, Jenny) benefit from your analysis and marketing strategy?
3. Are there any downsides, either to the company or to the customer, from using your analysis and marketing strategy? If so, how would you change your strategy?

Notes:

This scenario is not hypothetical. It is drawn from Charles Duhigg, "How Companies Learn Your Secrets," The New York Times Magazine, February 16, 2012, which reports how Target's analytics team built a pregnancy-prediction score from exactly this kind of purchase data. The article is worth reading in full.

Class Example 1.3 - UWL Institutional Data

UWL tracks data on every student, including demographics, majors, course completion, grades, retention, and graduation.

Why would they do this? What are the goals of UWL as an institution?

Class Example 1.4 - UWL Retention Disparities

Student data from previous years showed alarming relationships between retention rates and student race, socioeconomic status, and first-generation status.

Specifically, non-white students, students from lower socioeconomic backgrounds, and first-generation students were less likely to re-enroll in their second year at UWL.

Based on the goals of UWL, what options could UWL consider to address these disparities? What can you identify as the costs of those options?

Notes:

4. Summary: with great power comes great responsibility

- The ethical implications of data science are profound and pervasive.
- Data is **not** neutral
- Algorithms are **not** neutral

The fact that we, as data scientists, can potentially extract maximum information or yield the "best" predictions from data does not mean we should do so.

You used the following checklist in Activity 1.3. Here it is again, as one structure for considering the ethical aspects of any data science project:

- **Consent:** Did people meaningfully agree to this use of their data?
- **Clarity:** Is it transparent what is collected and why?
- **Consistency:** Are all people treated alike across groups?
- **Control & Transparency:** Can people see or correct what is held about them?
- **Consequences & Harm:** Who could be hurt, and how seriously?

We will return to these criteria throughout the course. They are not a checklist to complete once and file away.

5. Looking ahead

Everything in this course hangs on one repeating loop. You will see it in every module from here on:

```
ASK  ->  GET  ->  EXPLORE  ->  MODEL  ->  COMMUNICATE
 ^                                          |
 |________ findings raise new questions ______|
```

- **ASK**: frame the question.
- **GET**: acquire and inspect the data.
- **EXPLORE**: clean and contextualize, turning data into information.
- **MODEL**: find and test patterns.
- **COMMUNICATE**: interpret and share the knowledge so someone can act on it.

Notice where this module lives. Activity 1.5 and Class Example 1.2 were both **ASK** problems: what are we actually trying to find out, and should we be asking it at all? Module 2 takes up **GET**.

Real data science is cyclical, not a straight line.

Suggested Exercises

1. Consider the following three tasks:

   - A grocery chain computes last quarter's total revenue for each of its stores.
   - A streaming service decides which of 40 possible thumbnail images to show you for the same movie, based on which images people like you have clicked before.
   - A researcher writes a program to convert 50,000 scanned handwritten survey forms into a spreadsheet.

   a. Classify each as data science, statistics, software engineering, or some combination.

   b. For each, say which of the three features (learning from data, scale, a decision or insight as output) is present and which is missing.

   c. One of these is a *step inside* a data science project rather than a project itself. Which one, and which step of the ASK, GET, EXPLORE, MODEL, COMMUNICATE loop is it?

2. A city government hires a team to build a tool predicting which residential buildings are most likely to have a fire in the next year, so inspectors can be sent there first. The team consists of four software engineers and two statisticians.

   a. Which of Conway's three circles is the team missing?

   b. Give one specific way the missing circle could cause the tool to fail. Describe a mechanism, not a general worry.

   c. Suppose the tool works and inspections are reallocated. Name one group that could be harmed, and say which of the 5 Cs is most relevant.

3. A university proposes to install a system that uses the campus wireless network to record which buildings each student is in throughout the day. The stated purpose is to improve scheduling of building maintenance and computer lab hours.

   a. Apply the 5-C checklist. One sentence and a risk level (Low / Medium / High) for each criterion.

   b. The proposal notes that data will be "anonymized" by replacing student ID numbers with random codes. Does this change your assessment of the Consequences and Harm criterion? Explain, keeping in mind what a term-long record of one person's building visits reveals even without a name attached.

   c. Suggest one change to the proposal that would meaningfully improve its weakest criterion. State what the change would cost the university.

4. Return to the definition we adopted:

Data science is the art (or practice) of gaining, interpreting, and communicating insights from complex data through digital techniques.

a. Name one activity that satisfies this definition but that you would *not* call data science. What is the definition missing?

b. Name one activity you would call data science that this definition arguably excludes.

c. Propose a one-sentence revision. You are not expected to produce a perfect definition; you are expected to be able to say what yours gains and what it gives up.